# THE COMPLETED NAIVE COTANGENT COMPLEX IS WELL DEFINED IN K(B)

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*Editorial note.* Counted span is the author's own proof environment, source lines 512-557, following the statement at 505-510.

*Editorial note.* No omissions flagged; the closing 'and compute' refers to the one-line verification that the explicitly given h is a contracting homotopy.

*Editorial note.* The source typo 'morphsms' (line 545) is preserved verbatim.

*Allowed standard prerequisites (editorial).*

- Presentation-based construction of the completed NL. Section section-naive-cotangent-complex and equation equation-NL (same chapter, lines ~480-495). Defines NL from a presentation B = A[x]^wedge/J as (J/J^2 -> direct-sum B dx_i); the lemma shows independence of the presentation.
- Homotopy invariance for the algebraic NL. Algebra, Lemma algebra-lemma-NL-homotopy (cross-chapter, cited at ~520-521). The proof argues exactly as in this lemma; supplies the template for Steps 1 and 2.

*Source reference.* Original reference 00S1: focused-seeds/geometry/S9-restricted-nl-up-to-homotopy/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $A$ be a Noetherian ring and let $I \subset A$ be an ideal. Let $B$ be an object of (0AL4). The naive cotangent complex $NL^{\wedge}_{B/A}$ is well defined in $K(B)$.*

*Proof.* The lemma signifies that given a second presentation $B = A[y_1, \ldots, y_s]^{\wedge}/K$ the complexes of $B$-modules

$$(J/J^2 \to B\mathrm{d}x_i) \quad \text{and} \quad (K/K^2 \to \bigoplus B\mathrm{d}y_j)$$

are homotopy equivalent. To see this, we can argue exactly as in the proof of Algebra, Lemma 00S1.

Step 1. If we choose $g_i(y_1, \ldots, y_s) \in A[y_1, \ldots, y_s]^{\wedge}$ mapping to the image of $x_i$ in $B$, then we obtain a (unique) continuous $A$-algebra homomorphism

$$A[x_1, \ldots, x_r]^{\wedge} \to A[y_1, \ldots, y_s]^{\wedge}, \quad x_i \mapsto g_i(y_1, \ldots, y_s)$$

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compatible with the given surjections to $B$. Such a map is called a morphism of presentations. It induces a map from $J$ into $K$ and hence induces a $B$-module map $J/J^2 \to K/K^2$. Sending $\mathrm{d}x_i$ to $\sum(\partial g_i/\partial y_j)\mathrm{d}y_j$ we obtain a map of complexes

$$(J/J^2 \to \bigoplus B\mathrm{d}x_i) \longrightarrow (K/K^2 \to \bigoplus B\mathrm{d}y_j)$$

Of course we can do the same thing with the roles of the two presentations exchanged to get a map of complexes in the other direction.

Step 2. The construction above is compatible with compositions of morphsms of presentations. Hence to finish the proof it suffices to show: given $g_i(x_1, \ldots, x_r) \in A[x_1, \ldots, x_n]^\wedge$ mapping to the image of $x_i$ in $B$, the induced map of complexes

$$(J/J^2 \to \bigoplus B\mathrm{d}x_i) \longrightarrow (J/J^2 \to \bigoplus B\mathrm{d}x_i)$$

is homotopic to the identity map. To see this consider the map $h : \bigoplus B\mathrm{d}x_i \to J/J^2$ given by the rule $\mathrm{d}x_i \mapsto g_i(x_1, \ldots, x_n) - x_i$ and compute. $\square$